WORKED SOLUTION

STEP II 2021, Question 11

Probability and Statistics • Conditional Probability Recurrence

STEP II 2021, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.5, *The Last Passenger's Seat*: https://beyondstep.keithmcnulty.org/ch12/airline-seat.html

The key idea

Condition on where T_1 sits. If T_1 takes their own seat S_1, everyone else is fine; if T_1 takes S_k, then $T_2, \ldots, T_{k-1}$ sit happily and T_k becomes a new "random chooser". The crucial observation is that T_k is then in exactly the same position as T_1 was, but in a smaller train, with the vacant seat S_1 playing the part of T_k's own seat. That turns the problem into a recurrence.

Solution

Throughout, "T_r is displaced" means T_r arrives to find S_r already taken. Only a displaced passenger (or T_1) ever chooses at random.

Part (i)

With $n = 2$, T_1 takes S_1 or S_2 with probability $\frac{1}{2}$ each, and T_2 gets S_2 exactly when T_1 took S_1. So

$$P_2 = \tfrac{1}{2}.$$

With $n = 3$, condition on T_1's seat, each with probability $\frac{1}{3}$:

- S_1: T_2 and T_3 both find their seats free, so T_3 gets S_3.
- S_3: T_3 certainly does not get S_3.
- S_2: T_2 is displaced and chooses between S_1 and S_3 at random; T_3 gets S_3 exactly when T_2 picks S_1, with probability $\frac{1}{2}$.

Hence

$$P_3 = \tfrac{1}{3}\cdot 1 + \tfrac{1}{3}\cdot 0 + \tfrac{1}{3}\cdot \tfrac{1}{2} = \tfrac{1}{2}.$$

Part (ii)

Suppose T_1 sits in S_k with $2 \le k \le n-1$. Then $T_2, \ldots, T_{k-1}$ all find their seats free and take them. When T_k arrives, the free seats are

$$S_1,\ S_{k+1},\ S_{k+2},\ \ldots,\ S_n,$$

which is $n-k+1$ seats, and T_k chooses among them uniformly at random. The passengers still to come are $T_{k+1}, \ldots, T_n$, each wanting their own seat, and nobody still to come is allocated S_1.

? **Ask yourself** Why is this exactly the original problem with $n-k+1$ seats?

Relabel: call T_k the "first passenger" and pretend S_1 is T_k's allocated seat. In the original problem with m seats, the first passenger chooses uniformly from all m seats, each later passenger takes their own seat if free and otherwise chooses uniformly from the free ones, and the first passenger's allocated seat is wanted by no one else. Here, with $m = n-k+1$, the passengers $T_k, T_{k+1}, \ldots, T_n$ follow precisely those rules with the seats $S_1, S_{k+1}, \ldots, S_n$, and S_1 is wanted by no one else. The rules never use the seat labels except to decide whether a passenger's own seat is free, so the two processes have identical probabilities. In the smaller problem T_n is still the last passenger, so

$$\mathrm{P}(T_n \text{ sits in } S_n \mid T_1 \text{ sits in } S_k) = P_{n-k+1}.$$

Now, for $n \ge 3$, condition on T_1's seat, each of $S_1, \ldots, S_n$ having probability $\frac{1}{n}$. If it is S_1, T_n certainly gets S_n; if it is S_n, T_n certainly does not. So, by the law of total probability,

$$P_n = \frac{1}{n}\cdot 1 + \frac{1}{n}\sum_{k=2}^{n-1} P_{n-k+1} + \frac{1}{n}\cdot 0 = \frac{1}{n}\left(1 + \sum_{r=2}^{n-1} P_r\right),$$

where we substituted $r = n-k+1$, which runs from $n-1$ down to 2 as k runs from 2 to $n-1$. This is the required result.

Part (iii)

We claim that $P_n = \frac{1}{2}$ for every $n \ge 2$. We use strong induction.

Base case. $P_2 = \frac{1}{2}$ from part (i).

Inductive step. Let $n \ge 3$ and suppose $P_r = \frac{1}{2}$ for $r = 2, 3, \ldots, n-1$. The sum in part (ii) has $n-2$ terms, each equal to $\frac{1}{2}$, so

$$P_n = \frac{1}{n}\left(1 + \frac{n-2}{2}\right) = \frac{1}{n}\cdot\frac{n}{2} = \frac{1}{2}.$$

Hence $P_r = \frac{1}{2}$ for $r = 2, \ldots, n$, and by induction $P_n = \frac{1}{2}$ for all $n \ge 2$. (Part (i)'s $P_3 = \frac{1}{2}$ agrees.)

Part (iv)

For $n = 2$, $T_{n-1} = T_1$, who sits in S_1 with probability $\frac{1}{2}$. So $Q_2 = \frac{1}{2}$.

Now let $n \geq 3$, so T_{n-1} is not T_1, and again condition on T_1's seat S_k:

- $k = 1$: everyone gets their own seat, so T_{n-1} gets S_{n-1}.
- $k = n$: $T_2, \ldots, T_{n-1}$ all find their seats free, so again T_{n-1} gets S_{n-1}.
- $k = n - 1$: T_{n-1} certainly does not get S_{n-1}.
- $2 \leq k \leq n - 2$: exactly as in part (ii), the rest of the journey is the original problem with $m = n - k + 1$ seats, led by T_k. In that smaller train T_{n-1} is the second-to-last passenger, so the probability is Q_{n-k+1}. Here m runs over $3, 4, \ldots, n - 1$.

Therefore, for $n \geq 3$,

$$Q_n = \frac{1}{n}\left(2 + \sum_{m=3}^{n-1} Q_m\right).$$

For $n = 3$ the sum is empty and $Q_3 = \frac{2}{3}$. If $Q_m = \frac{2}{3}$ for $3 \leq m \leq n - 1$, then

$$Q_n = \frac{1}{n}\left(2 + \frac{2(n-3)}{3}\right) = \frac{1}{n} \cdot \frac{2n}{3} = \frac{2}{3},$$

so by strong induction $Q_n = \frac{2}{3}$ for every $n \geq 3$. In summary,

$$Q_2 = \tfrac{1}{2}, \qquad Q_n = \tfrac{2}{3} \quad (n \geq 3).$$

? **Ask yourself** Is there a way to see $\frac{1}{2}$ and $\frac{2}{3}$ without any recurrence?

Yes. Whenever someone chooses at random, S_1 and each still-free seat of a later passenger are treated alike, so until the process ends the seats S_1 and S_n are equally likely to be the one taken first; T_n succeeds exactly when S_1 goes first, giving $\frac{1}{2}$. For T_{n-1} (with $n \geq 3$), look at which of S_1, S_{n-1}, S_n is taken first by a random chooser; each is equally likely, and T_{n-1} is displaced only if it is S_{n-1}, giving $1 - \frac{1}{3} = \frac{2}{3}$.

Checks

- The case $n = 3$ can be listed by hand: the outcomes for (T_1, T_2, T_3) are (S_1, S_2, S_3) with probability $\frac{1}{3}$, (S_3, S_2, S_1) with probability $\frac{1}{3}$, and (S_2, S_1, S_3), (S_2, S_3, S_1) with probability $\frac{1}{6}$ each. So T_3 gets S_3 with probability $\frac{1}{2}$ and T_2 gets S_2 with probability $\frac{2}{3}$, as found.
- Exact enumeration of every possible journey by computer for $n = 2, \ldots, 8$ gives $P_n = \frac{1}{2}$ throughout, $Q_n = \frac{2}{3}$ for $n \geq 3$, and confirms the conditional probability in part (ii) for every k. A simulation of 10^5 journeys with $n = 20$ gave 0.500 and 0.668.

Common slips

- Forgetting that T_1 can choose S_1 itself, which gives the "$1+$" in the recurrence.
- In part (iv), missing the case where T_1 takes S_n: this is good news for T_{n-1}, so the constant becomes 2, not 1.
- Applying the Q recurrence at $n = 2$, where T_{n-1} is T_1 and the answer is $\frac{1}{2}$, not $\frac{2}{3}$.

Beyond STEP

The recurrence here is a first-step analysis: condition on the first random choice and recognise a smaller copy of the same problem. The same technique gives hitting probabilities and expected hitting times for Markov chains, a central topic in first-year probability. The symmetry argument above is a taste of how a well-chosen "which happens first?" question can replace a long calculation. Grimmett and Welsh, *Probability: An Introduction*, develops both ideas.

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